

Photoionization cross section and asymmetry parameter of LiH: a mixed GTO/STOCOS L^2 basis set calculation

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Received 15 February 1996

Abstract

The outer-shell photoionization cross section and the asymmetry parameter are computed ab initio for LiH up to a photon energy of 50 eV. The calculations employ an L^2 K-matrix technique at the Static-Exchange Approximation (SEA) and at the Random Phase Approximation (RPA) levels, using mixed GTO/STOCOS bases. The results show a fairly good internal coherence and are in good agreement with those obtained previously with pure Polynomial Spherical GTO (PSGTO) bases.

1. Introduction

The recent advances in the experimental study of the photoionization of small molecules, notably the possibility to measure accurately the differential cross sections, have produced a great interest in the theoretical methods for calculating these quantities. While the calculation of the integral photoionization cross sections for small molecules is becoming a well understood field, very few ab initio calculations of the asymmetry parameters exist to date. Indeed, the total cross sections may be satisfactorily obtained from relatively simple calculations, particularly when the cross section does not present a structure due to autoionizing states. In such cases, the Stieltjes imaging technique allows to extract these informations from the spectral momenta of the oscillator strength density, which do

not necessitate a refined representation of the continuum [1]. On the contrary, the determination of the asymmetry parameters requires a detailed knowledge of the continuum wavefunctions, including the phase-shifts of their outgoing components. Among the several techniques developed to this purpose, it is worth to recall the Schwinger variational method [2], the continuum multiple scattering method [3], the linear algebraic method [4] and the complex basis function method [5].

The K-matrix method upon L^2 basis that our group has developed in the last years [6–10] appears to be a suitable technique for the calculation of accurate integral and differential cross sections in molecular systems.

Formally, the K-matrix technique for the wavefunctions is a variational method that expands the continuum eigenstates at a given energy upon one or more sets of continuum ‘unperturbed’ basis functions with a smooth energy dependence and a set of localized states. The former represent the background contin-

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uum and are spin- and symmetry-coupled products of the ion states times ‘partial waves’. The localized functions, instead, allow to account efficiently for any short-range feature: bound states, resonances, short-range deviations of the true potential from that determining the ‘unperturbed’ continuum eigenstates and correlation effects. The method presents two appealing features for molecular calculations. First, the continuum expansion may be efficiently represented by a quadrature upon a finite L^2 basis and this brings the method into the usual computational framework of theoretical chemistry. In addition, the formalism is easily adapted to compute the RPA amplitudes upon the corresponding bases [6]. This represents a nice feature of the method, since the RPA is well known to provide a substantial improvement of the transition matrix elements when the wavefunctions are not or poorly correlated.

Our code, therefore, should fit rather straightforwardly in the existing packages for molecular calculations and thus allow to exploit their computational power with moderate additional effort.

The applicability of the method rests therefore on a suitable L^2 representation of the ‘partial waves’ and this, in turn, poses severe requirements on the basis orbitals. As explained below, the partial waves may be considered the eigenstates with given l, m values of a single particle in a suitable projection of the ionic potential in which the l, m channels are decoupled. Thus the orbital basis required by a K-matrix calculation must be able to represent the occupied orbitals of the system, the lowest virtual orbitals, eventual shape resonances and the partial waves for the continuum. These last must be represented at least in the whole ‘molecular region’, i.e. up to a distance where the potential may be considered Coulombic so that the waves may be continued to infinity as shifted Coulomb ones.

For atomic systems the most suitable bases, as used by our group, include STO, HO (Hydrogenic Orbitals) and STOCOS (STO times a Cosine) [6]. Applied to non-trivial molecular systems, a pure STO/STOCOS one-center basis requires an inordinately large computational effort to give really accurate results. In fact, one would need a huge basis set to reproduce accurately the functions near the off-center nuclei; the method appears hardly viable when these nuclei are more than protons and indeed such expansions are generally performed only for the hy-

drides. Compared with the more standard calculations of the lowest bound states at the same accuracy level, one needs larger orbital bases with rather redundant metrics and hence more accurate matrix elements. At the present state-of-art, this rules out a multi-center STO/STOCOS expansion because, in spite of a large research effort [11], the bielectronic integrals may be computed with the necessary accuracy only by multidimensional numerical integrations that are prohibitively expensive.

In previous works, therefore, our group has explored the possibility to employ gaussian bases in K-matrix calculations, first in atomic systems (He, Mg) [8] and then in a few light molecules (H_2 , Li_2 , LiH) [9,10]. It is not superfluous to recall that the phase shifts of the GTO-expanded partial waves cannot be obtained from the analysis of their asymptotic behaviour, as possible with the STOCOS functions, but may be obtained from a preliminary single-channel K-matrix calculation [8].

Previously [10] polynomials in r times a spherical GTO, called Polynomial Spherical GTO (PSGTO) were employed. They proved considerably more suitable than the simple GTO but lead to more expensive integrals. While the results so obtained proved accurate, the capability of a PSGTO basis to span an extended molecular region (and thus the applicability of the method to medium-sized molecules) may impose very demanding requirements upon the polynomial employed. The relative smallness of the spanned region, moreover, is reflected also in the energy distribution of the ‘unperturbed’ waves yielded by a PSGTO basis. About half of the states, indeed, are bound ones or have very low wave energies, while in the most interesting energy range their spacing grows with extreme rapidity. As an example, the Li-centered s-wave spectrum employed in Ref. [10] contained 11 bound states, 6 waves in the first 0.15 au above the threshold, other 5 waves between the wave energies 0.56 and 1.70 au and finally a few states at very high energy which cannot represent the sought for continuum. Such a distribution represents an obvious weakness unless there is evidence that at medium and high wave energies the partial waves properties and the matrix elements among them may be safely interpolated upon these few points. It is almost needless to say that high wave energies are necessarily required when one studies the interaction of ionization channels with largely

different thresholds.

A clever choice of polynomial gaussian orbitals and possibly the use of quadruple precision computer packages might considerably enlarge the applicability field, but bases more capable to reproduce the continuum partial waves appear highly desirable.

Considering both the inherent limitations and the technical difficulties, it seems fair to say that a pure GTO/PSGTO or STO/STOCOS basis is unlikely to represent the optimal choice for a molecular calculation and thus it appeared worth to investigate mixed bases in which the one-center components (i.e. the partial waves) are represented by STO and STOCOS, while the many-center ones are represented by GTO. This approach requires, of course, a suitably accurate and efficient calculation of the one- and two-electron integrals between one or two one-center STOCOS and arbitrarily located GTO. At the present, no satisfactory analytical approximation is available for computing these integrals, so in this work we have computed them by projecting the STOCOS upon relatively small GTO bases (20–35 for each l, m symmetry). This approach, perhaps best known as spectral representation of the operators, has been introduced in molecular calculations by Huzinaga and coworkers to evaluate the Coulomb and exchange integrals [12] and was occasionally employed to compute the two-electron integrals between many-center STO [11]. This approach appears promising since requires only matrix manipulations (easily performed on modern computers), the well-established calculation of two-electron integrals between GTO and the evaluation of GTO/STOCOS overlaps. These last integrals, at the present rather non-standard, do not represent a time-determining step because their limited number allows an efficient calculation also without fully optimized routines.

2. Method of calculation

For all the aspects not directly related to the basis orbitals, the present calculations are quite similar to those performed previously by our group using PSGTO bases [10]. In particular, the underlying theory has been described in detail elsewhere [6] and thus only a brief outline is given here.

The present calculations are performed at single-channel single-configuration level, using both the

static-exchange (SEA) and random phase (RPA) approximations. In the energy range considered (up to a photon energy of 50 eV), the channel couplings are negligible and therefore the excitations from the 1σ orbital may be safely neglected. The vibrational structure of the states has been neglected and the rotation has been treated classically, i.e. the integral and differential cross sections have been averaged upon the orientations.

The ground state has been described by an SCF function on the multicenter GTO basis described in the previous PSGTO calculation [10], essentially a 6-311G set augmented with five diffuse GTO to represent the lowest virtual orbitals. Being the molecular symmetry $C_{\infty v}$, all the symmetries $l \geq 0, m = 0$ contribute to the expansion of the σ waves and all the symmetries $l \geq 1, m = \pm 1$ to that of the π waves. The expansion is here truncated at $l = 8$; an exploratory calculation truncated at $l = 7$ has yielded nearly coincident results.

2.1. The bases for the partial wave channels (pwc)

Each formal basis set employed to describe the continuum is the spin-coupled direct product of the desired ion state times a one-electron manifold that asymptotically behaves as a spherical shifted standing Coulomb wave. These sets are therefore called 'partial wave channels' (pwc) and denoted by a greek index that specifies collectively the ion state $|\Phi_j^+\rangle$ with energy E_j^+ and the l, m values of the wave, i.e. $\beta = (j_\beta, l_\beta, m_\beta)$. The basis functions $|\Phi_{\beta, E}\rangle = |\Phi_j^+ \phi_{j, l, m, \epsilon}\rangle$ of the pwc will be represented upon a finite L^2 basis and therefore their properties must depend smoothly upon the energy index $E = E_j^+ + \epsilon$. An efficient way to construct a set of L^2 waves $|\phi_{j, l, m, \epsilon}\rangle$ with these properties is to diagonalize upon a suitable set of one-center long-range orbitals a zeroth order Hamiltonian of the form

$$\hat{h}_j^0 = \sum_{l, m} \hat{h}_{jlm}^0; \quad \hat{h}_{jlm}^0 = |Y_{lm}\rangle \langle Y_{lm}| \hat{h}_j |Y_{lm}\rangle \langle Y_{lm}|,$$

where $\hat{h}_j$ is the one-electron SEA Hamiltonian describing the motion of one electron in the ionic field corresponding to the hole in the ground state occupied orbital j . The waves so obtained are clearly non-

orthogonal to the occupied orbitals and the matrix elements must be evaluated accordingly.

Using only these basis functions would yield little more than a one-center expansion, whose shortcomings in molecular calculations are well known. Therefore the basis includes also a set of localized functions, referred to as 'localized channel' (lc), in which the outer orbitals are expanded upon a multicenter gaussian basis: this allows to reproduce accurately the short-range features of the wavefunctions. These orbitals were obtained by diagonalizing the SEA Hamiltonian on the basis of the SCF virtual orbitals.

The bases employed to represent the continua are made of STO and STOCOS functions. These last have the form [6]

$$\chi_{\zeta,k,n,l,m}(\mathbf{r}) = N_{\zeta,k,n} r^{n-1} e^{-\zeta r} \cos(kr) Y_{l,m}(\theta, \phi).$$

It is convenient to choose $n = l + 1$ and the k values according to $k_j = \pi(j + \frac{1}{2})/R_0$ so that all these orbitals have a radial node in $r = R_0$. With a proper choice of the parameters ζ and R_0 , the diagonalization of the SEA effective Hamiltonian upon a basis of this kind yields a large number of closely spaced variational wavepackets that are almost exact up to about R_0 , i.e. inside this sphere they may be considered proportional to the true continuum eigenstates of the same wave energy. In the present work $R_0 = 100 a_0$ and the proportionality constant and the phase shift are determined from a comparison with a shifted energy-normalized Coulomb wave between 40 and 70 a_0 . Each pwc contained about 70–90 states, of which 60–80 lie in the interesting continuum region; their energy spacing increases steadily from about 0.01 au at the threshold to 0.1 au at the upper end of the region considered (a wave energy of about 2 au). The total dimension of the K-matrix calculation was about 700 for the $m = 1$ waves and 800 for the $m = 0$ ones.

At RPA level, the transition matrix elements between the ground state $|\Psi_0\rangle$ and a state $|\Psi_{\alpha,E}^{(-)}\rangle$ at the continuum energy E and asymptotically behaving as the product of the ion state j_α times an outgoing α -partial wave (plus ingoing waves in all channels) is given by [6]

$$\begin{aligned} \langle \Psi_0 | \hat{O} | \Psi_{\alpha,E}^{(-)} \rangle &= \sum_{\gamma} \left\{ \langle \Psi_0 | \hat{O} | \Phi_{\gamma E} \rangle \right. \\ &+ \sum_{\beta} \oint d\epsilon \left[\langle \Psi_0 | \hat{O} | \Phi_{\beta\epsilon} \rangle \frac{\mathcal{P}}{E - \epsilon} \mathbf{K}_{\beta\epsilon,\gamma E} \right. \\ &\left. \left. + \langle \Phi_{\beta\epsilon} | \hat{O} | \Psi_0 \rangle \mathbf{L}_{\beta\epsilon,\gamma E} \right] \right\} [\mathbf{1} - i\pi \mathbf{K}(E)]_{\gamma,\alpha}^{-1}, \end{aligned}$$

where $\mathcal{P}$ is the principal part of the integral, α and γ represent open channels and β an arbitrary one. $\mathbf{K}$ and $\mathbf{L}$ are smoothly energy-dependent expansion coefficients, related to the standard RPA matrices $\mathbf{X}$, $\mathbf{Y}$ by

$$\mathbf{X}_{\beta\epsilon,\alpha E} = \delta_{\alpha\beta} \delta(\epsilon, E) + \frac{\mathcal{P}}{E - \epsilon} \mathbf{K}_{\beta\epsilon,\alpha E},$$

$$\mathbf{Y}_{\beta\epsilon,\alpha E} = \mathbf{L}_{\beta\epsilon,\alpha E}.$$

The SEA expression for the transition matrix elements may be obtained by simply dropping $\mathbf{L}$. The equations determining these coefficients at both levels are discussed in a previous review [6].

The total cross section is immediately obtained from the photon energy and the squares of these matrix elements. The asymmetry parameter β requires not only these matrix elements but also the phaseshifts of the unperturbed waves $|\Phi_{\gamma E}\rangle$.

3. Computational details

A full description of the method employed to compute the needed integrals will be given in a forthcoming article, so only the essential points are outlined here.

3.1. The matrix elements

With the present choice of the N -electron basis, the SEA/RPA matrix elements require the calculations of one-electron integrals of types 2S/0G (i.e. among 2 STO/STOCOS and no GTO), 1S/1G and 0S/2G and two-electron integrals of types 2S/2G, 1S/3G and 0S/4G. No long-range integrals involving three or more STOCOS are required.

The strategy underlying the calculation of the mixed integrals (i.e. those involving both STOCOS and GTO) is to project the STO/STOCOS upon bases

of GTO capable of representing them accurately in the regions of space that yield the main contribution to the integrals. The effort and the accuracy loss of these projections may become quite heavy when the integrands are large in very extended regions of space. The standard trick to overcome these difficulties is to subtract from the integrands a function whose integrals may be evaluated analytically and that has a similar asymptotic behaviour.

The integrals needed by the matrix $\mathbf{Q}$ in the RPA equations are essentially short-range exchange integrals and need no specific discussion. The most delicate integrals arise from the matrix elements of the SEA effective Hamiltonian on the outer orbitals (STOCOS S_i and GTO G_i), which form the matrix $\mathbf{M}$ of the RPA equations [6] and are computed as

$$\begin{aligned}\langle S_a | h_{eff} | S_b \rangle &= \langle S_a | T + V_N + J_A | S_b \rangle \\ &+ \langle S_a P_a | (J - J_A) + K | P_b S_b \rangle, \\ \langle S_a | h_{eff} | G_b \rangle &= \langle S_a P_a | T + V_N + J + K | G_b \rangle\end{aligned}$$

As usual, T is the kinetic energy, V_N the nuclear potential, J and K the Coulomb and exchange operators; P_a and P_b represent the projectors for the pwc to which the STOCOS belong, while J_A is a potential that behaves asymptotically as J and thus makes $J - J_A$ a short-range potential.

Indeed, the projection of the far-propagating STOCOS upon the short-range GTO bases would be hopeless for computing directly the Coulomb integrals involving two STOCOS, because this potential contains a monopole component that at large distances damps as slowly as $1/R$.

The inclusion of J_A is not needed in the matrix elements $\langle S_a | h_{eff} | G_b \rangle$ since their integrands are clearly short-range.

Obviously, the optimal J_A should make $J - J_A$ short-range, without divergencies and (hopefully) as small as possible, because the integrals evaluated through the projection are clearly the least accurate among those involved in the calculations. It should not be forgotten that, to yield reliable waves, the STOCOS bases must be rather redundant (the smaller accepted metric eigenvalues are typically around or below 10^{-6}) and therefore the errors in these integrals may be enhanced during the transformation into an orthogonal basis that is performed to obtain the waves. Moreover, the time-

determining step is the calculation of the two-electron integrals between the GTO employed for the basis and for the projection: a proper choice of J_A should allow an accurate projection with the smallest possible basis and thus reduce the computational effort. Two forms of J_A have been examined in the present work: the electrostatic potential generated by a set of discrete point charges (J_{Ad}) and by a spherical gaussian charge distribution centered in the STOCOS origin (J_{Ag}).

The simplest choice for J_{Ad} is, of course, the fraction $-N_i/Z_t$ of the nuclear potential, where N_i is the number of electrons in the residual ion and Z_t the total nuclear charge. In the present calculations for LiH this choice proved fairly accurate in almost all the cases considered, with the only exception of the matrix elements involving two s waves centered on the lithium nucleus. In this case, in fact, $J - J_{Ad}$ weights heavily the imprecisions of the projection near the nucleus, where a GTO expansion reproduces with great difficulties the cusp of the s waves. A few tests suggested that this difficulty may be overcome by using an optimized set of point charges, but this procedure was not followed further because introduces undesirable parameters and is contrary to the sought for minimization of $J - J_A$.

These integrals were therefore evaluated with the asymptotic potential J_{Ag} , which in the present case was three times the electrostatic potential generated by an electron placed in one 1s GTO centered in the origin. The calculations were performed using several of the GTO already employed to expand the occupied MO, so that $\langle S_a P_a | J_{Ag} | P_b S_b \rangle$ did not need new two-electron integrals, and proved quite independent from the orbital exponent of the GTO actually employed. The analytical one-electron integrals $\langle S_a | J_{Ag} | S_b \rangle$, are rather expensive but also relatively few and thus their calculation is not a time-determining step.

For larger molecules, it appears fair to say that a combination of these two asymptotic potentials should ensure a sufficient accuracy, with the point charges employed preferentially to damp the eventual contributions from compact inner shells of off-center nuclei. Anyway, more refined and smoother forms appear desirable and are currently under study.

3.2. GTO/STOCOS overlaps

The STOCOS of each pwc were projected upon a separate GTO basis. These bases were enlarged until the properties calculated for the pwc went to convergence. Their composition depends on the symmetry of the wave and the position of the expansion center, but all these bases share a few common features. In all cases we employed a set of GTO on the STOCOS center with the same l, m and a set of GTO with the lowest possible l (i.e. $l = |m|$) on the other nucleus. Both sets of GTO had the orbital exponents chosen in geometrical progression of ratio around 1.5. Each set contains from 10 to 20 GTO and the total dimension for each pwc was never greater than 35.

The transition matrix elements, in both the velocity and the length gauges, may be reduced to sums of overlap integrals between GTO and STOCOS/STOSIN times angular factors. These matrix elements were computed also by the projection technique and the results proved always in very good agreement.

The analytical matrix elements $\langle S_a | J_{Ag} | S_b \rangle$ required by the asymptotic potential J_{Ag} may be reduced to one-center overlap integrals between GTO and STOCOS/STOSIN.

The one-center overlaps may be computed analytically, although the quadruple precision is mandatory to obtain reliable results. The two-center overlaps were reduced to one dimensional integrations by expanding the gaussian orbital upon the STOCOS origin and hence evaluated numerically through an adaptive numerical integration.

3.3. Sample values

Although a detailed analysis of the accuracy of these integrals will be the object of a forthcoming paper, it seems useful to discuss here a few cases by comparing the matrix elements calculated in this way and by numerical integrations of J and K . We will consider only integrals involving $m = 0$ STOCOS, because these are the most difficult to evaluate by a finite representation (either on projection basis or a numerical grid). For the $m = 1$ orbitals, indeed, the presence of a nodal plane in the molecular axis removes the most critical contributions to the integral. The absence of these contributions is also reflected in the smallness of the phase shifts of the π waves with respect to the σ ones.

The GTO/STOCOS one-center overlaps are analytical and may be considered exact. Moderate relative errors occur only when the absolute values are very small. The routines for the two-center overlaps, that in our code are reduced to one-dimensional quadratures involving Bessel functions, have been checked against two-dimensional numerical integrations. In almost all cases the results computed by these two methods were identical at least to ten significant digits, therefore these integrals may be considered exact if compared with the other approximations involved. In Table 1, we report the largest differences found in a cross-check of the two integrations, which sampled all the parameter range involved in the present calculations. The accuracy of these overlap integrals is a necessary condition for that of the final matrix elements, which requires also an 'effective completeness' of the projection basis and the numerical stability of the matrix operations. In Table 2 we compare the matrix elements for three representative Li-centered $l = 0$ orbitals as computed with two different methods; also these matrix elements are among the more critical encountered in the present work. The projection basis used for this wave is built by 20 GTO centered on the lithium with exponents ranging from 0.01 to 22.2 in a geometric progression of ratio 1.5 and 10 GTO centered on H with exponents from 0.1 to 6.87 with ratio 1.6.

The first part of Table 2 compares the elements of the SEA effective Hamiltonian as computed in the present work using J_{Ag} (left) and by a five-dimensional numerical quadrature of $J + K$ (right, $T + V$ is analytical). The agreement is very good for the matrix elements involving at least one localized valence orbital, while is less satisfactory among two STOCOS. In these cases, however, the estimated error of the numerical integration is larger than the difference with the projected results. It must be noted that this projection basis is unable to yield an accurate matrix using J_{Ad} , which requires more concentrated GTO.

The matrix Q in the RPA equations [6] requires essentially the exchange integrals among the valence orbitals; these integrals are therefore reported in the second part of Table 2. Since the integrands are short-range and without divergencies, the two methods yields very close results.

In comparing these quantities, one should not forget that the GTO basis is unable to provide a long-range

Table 1

Comparison of the GTO/STOCOS overlaps computed by the method employed in this work with the results of two-dimensional quadratures. The STOCOS parameters are defined in Eq. (1), with $n = l + 1$; the GTO have the form $r^{l'} \exp(-\alpha r^2) Y_{l',m}(\theta, \phi)$ and have the center in (0, 0, 3). The last column is an estimate of the relative accuracy of the two-dimensional integration

l	m	ζ	k	l'	α	This work	Numerical	Accuracy
0	0	0.12	3	0	0.01	$-7.7518102164 \times 10^{-5}$	$-7.7518102071 \times 10^{-5}$	2×10^{-5}
1	0	0.12	1	0	10.00	$-5.6773084615 \times 10^{-3}$	$-5.6773084608 \times 10^{-3}$	1×10^{-9}
1	1	0.12	1	1	0.01	$1.3862850765 \times 10^{-3}$	$1.3862850768 \times 10^{-3}$	3×10^{-6}
1	1	0.12	1	1	1.00	$-8.4270577649 \times 10^{-3}$	$-8.4270577647 \times 10^{-3}$	7×10^{-9}
1	1	0.12	1	1	50.00	$-8.1648381194 \times 10^{-5}$	$-8.1648381196 \times 10^{-5}$	4×10^{-9}

Table 2

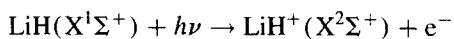
Matrices of a few operators upon three representative outer basis orbitals as computed in the present work and by numerical integration. All the orbitals are of 1s-type ($n = 1, l = 0$) and have (ζ, k) values (1, 0); (0.12, 0.9896) and (0.12, 1.4923); see Eq. (1)

SEA effective Hamiltonian						
present work (see text)				numerical $J + K$		
-1.290137				-1.290047		
-0.034164		0.380449		-0.034297		0.379866
-0.001865		-0.045086	0.999947	-0.001955		-0.045464
				0.996112		
Exchange						
projected				numerical		
-0.524418				-0.524313		
-0.058333		0.005457		-0.058270		0.005463
-0.020435		-0.000490	0.000880	-0.020306		-0.000483
				0.000883		
Overlap						
projected				analytical		
1.000000				1.000000		
-0.079595		0.674986		-0.079592		1.000000
-0.067757		-0.115419	0.434505	-0.067750		-0.078086
				1.000000		

representation of the STOCOS. This appears clearly in the last part of Table 2, that compares the STOCOS overlaps $\langle S_a | S_b \rangle$ with their projected counterparts $\langle S_a | P | S_b \rangle$: one of the STOCOS is less than half-projected on the GTO basis and the relatively large off-diagonal STOCOS/STOCOS overlap is in error by a factor 1.5. The projection method yields nevertheless accurate value for the exchange and (modified) Coulomb integrals, meaning that the projection basis provides an effective spectral representation of the potential [12].

4. Results

The process studied is



for photon energies up to 50 eV, where only the ionization channels corresponding to the ground state of the parent molecular ion $(2\sigma)^{-1}$ are open. Neglecting excitations from the 1σ core orbital, the dipole-allowed partial wave channels to be considered are $2\sigma \rightarrow \epsilon\sigma(^1\Sigma)$ and $2\sigma \rightarrow \epsilon\pi(^1\Pi)$.

The geometry (equilibrium experimental distance of 3.0154 bohr) and the SCF orbitals, both occupied

and virtual, were the same employed in the previous calculations with PSGTO bases [10], so the two sets of results are strictly comparable. The ionization potential is 8.203 eV, against the experimental estimate 7.7 eV.

In absence of reliable experimental data, the accuracy of the calculations may be judged only from their internal consistency and the agreement with other theoretical results; to our knowledge the only other calculations are the PSGTO ones of our group. On the whole, the present GTO/STOCOS results agree rather closely with the previous PSGTO ones, but display a greater regularity with respect to the previous ones. Unfortunately, the cross sections are rather featureless and thus do not provide a stringent comparison, but the agreement of the partial wave phaseshifts that will be discussed below is quite noticeable.

The internal consistency of the present and previous calculations is supported not only by the agreement of the RPA calculations in both the length and velocity gauges, but also by the very weak dependence of the results upon the expansion center of the STOCOS. In fact, we have performed two sets of calculations placing the STOCOS center on the Li and H nuclei respectively and the results are in good agreement. It should be noted that in the ion ground state the net charge is located essentially on the lithium, while the hydrogen is practically neutral, therefore the lithium represents a far better origin for the expansion in spherical waves of the continuum states. The H-centered calculations, therefore, were carried out essentially as a check of the Li-centered ones; the close agreement of the results obtained with these origins strongly suggests that either the PGSTO and GTO/STOCOS bases employed may be considered almost complete for the expansion of the electron waves in the molecular region, at least within the limits of the approximations employed (SEA, RPA). It should not be overlooked that this origin invariance involves the rather large translation of 3.0154 bohr and thus represents a not trivial check.

The GTO/STOCOS and PSGTO results are very close in almost all respects: the only noticeable differences are a slight variation of the Li-centered $s\sigma$ phaseshifts and an improvement of the low-energy results for the H-centered cross section. In this region, in fact, the PSGTO results showed small but evident oscillations that have practically disappeared in the present ones. All the other features discussed below

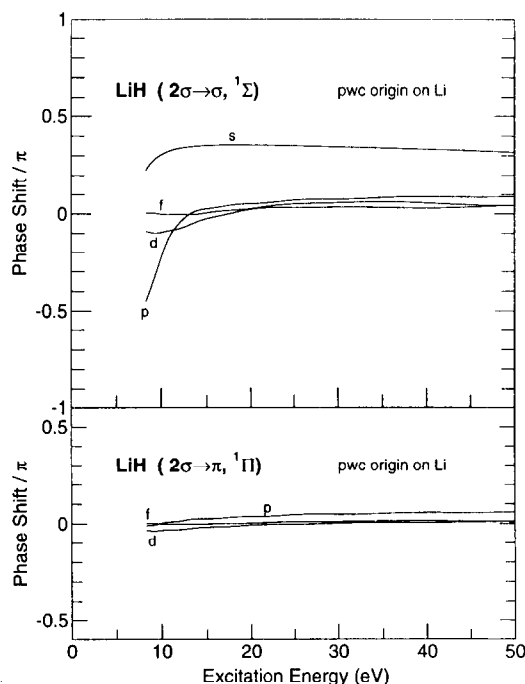


Fig. 1. Phase shifts of the partial wave channel orbitals with respect to pure Coulomb waves. The expansion center of the partial wave channels is on the Li atom. The results for both the $2\sigma \rightarrow \epsilon\sigma$ and $2\sigma \rightarrow \epsilon\pi$ partial ionization channels are shown. For higher values of l the phase shifts are negligible.

are common to both the PSGTO and GTO/STOCOS calculations.

The phase shifts of the pwc basis functions with the expansion center located upon the lithium and hydrogen atoms are reported in Figs. 1 and 2 respectively.

For $l \geq 3$, the phase shifts of the H-centered σ waves have a local maximum (in absolute value), whose energy increases with l . This feature is due to the centrifugal potential that acts as a barrier localized at the H nucleus and keeps the low-energy orbitals far from the potential well around the Li nucleus. All these shifts are very close to those obtained with the PSGTO bases: only the s-waves centered on the lithium atom show an appreciable difference, while all the other curves are practically coincident on these scales. Fig. 3 reports the phase shifts of the Li-centered $s\sigma$ ($l, m = 0, 0$) and $p\sigma$ ($l, m = 1, 0$) waves as computed with three different methods. The full lines are the present GTO/STOCOS results, the dashed lines those of the previous PSGTO calculations, while the

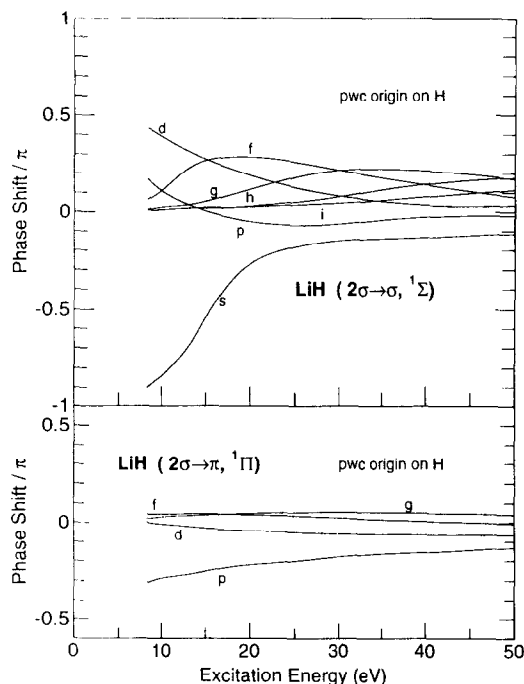


Fig. 2. Phase shifts of the partial wave channel orbitals with respect to pure Coulomb waves. The expansion center of the partial wave channels is on the H atom. The results for both the $2\sigma \rightarrow \epsilon\sigma$ and $2\sigma \rightarrow \epsilon\pi$ partial ionization channels are shown. For higher values of l the phase shifts are negligible.

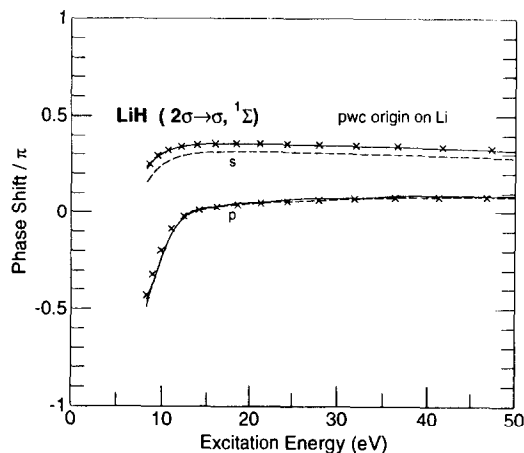


Fig. 3. Phase shifts of the $s\sigma$ ($l, m = 0, 0$) and $p\sigma$ ($l, m = 1, 0$) partial wave channel orbitals with respect to pure Coulomb waves. The expansion center of the partial wave channels is on the Li atom. Full line: present results; dashed line: PSGTO results [9]; $\times$: results of a one-center SCF/SEA calculation with pure STOCOS bases.

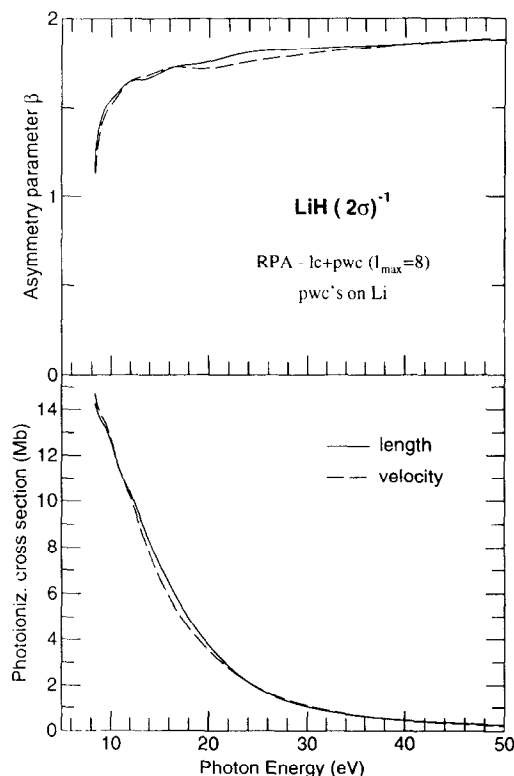


Fig. 4. Photoionization cross section and asymmetry parameter as a function of the photon energy. Expansion center for the partial wave channels on the Li atom. The basis includes the localized channel and the partial wave channels up to $l = 8$.

crosses are those of a one-center SEA calculation upon a pure STOCOS basis. The almost coincidence of the GTO/STOCOS and STOCOS results is gratifying and indicates that the electrostatic potential of LiH^+ on these waves is reproduced by a one-center expansion of the occupied orbitals better than one might expect. The difference for the $s\sigma$ waves might be due to a poor representation of the very small r region in the GTO calculation. Since these methods to compute the phaseshifts are completely independent, this agreement can by no means be considered fortuitous and confirms the accuracy of both the GTO/STOCOS and PSGTO calculations.

The total photoionization cross section and the asymmetry parameters computed at RPA level with the STOCOS origin on Li and H respectively are shown in Figs. 4 and 5. Not surprisingly, the results in the first case show a better gauge invariance and a smoother behaviour.

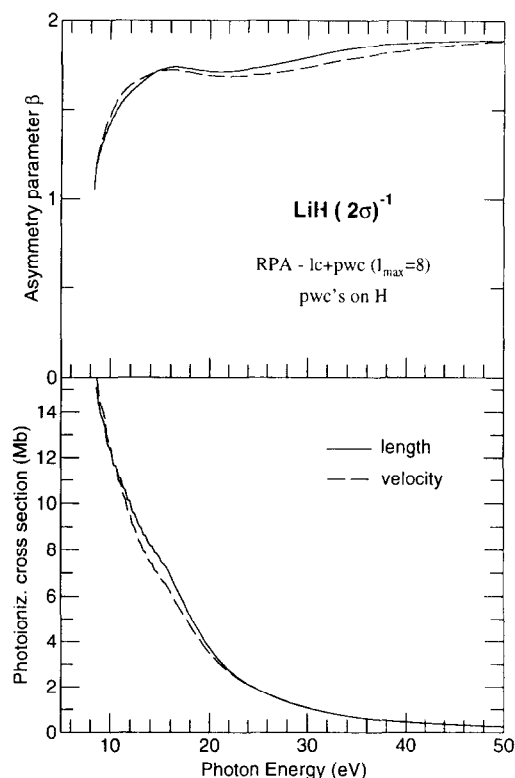


Fig. 5. Photoionization cross section and asymmetry parameter as a function of the photon energy. Expansion center for the partial wave channels on the H atom. The basis includes the localized channel and the partial wave channels up to $l = 8$.

While the gauge invariance at RPA level may be considered good, that at SEA level is rather poor, particularly at low energies. The mean of the SEA VG/LG results is nevertheless in good agreement with the RPA values. For the asymmetry parameter β , the SEA and RPA results are in better agreement than for the cross section.

The inclusion of the localized channel has limited effects when the expansion center is on the lithium, but is fundamental to obtain an accurate asymmetry parameter if this is on the hydrogen. The H-centered total cross section, instead, is weakly affected by the presence of these orbitals. Fig. 6 reports the RPA results obtained without the localized channel and the center on this atom. Quite evidently, a one-center expansion on the hydrogen cannot be truncated at $l = 8$, although for higher l only the short-range components appear needed. It is interesting to note that, due to the large internuclear distance, the localized channel af-

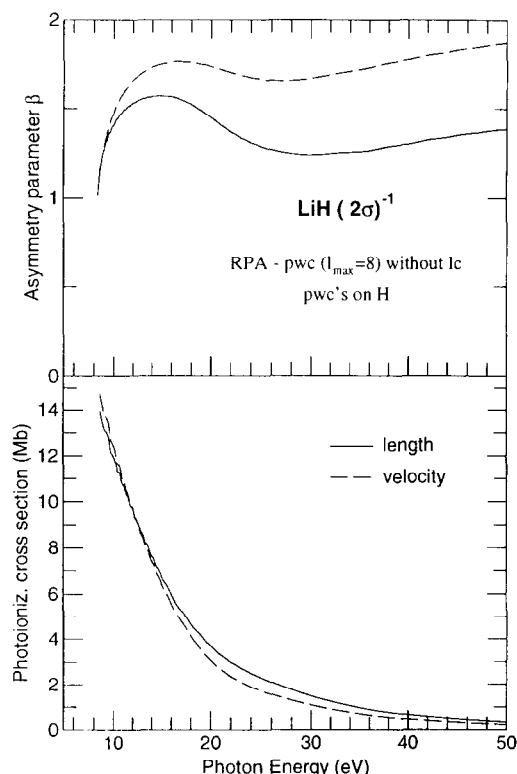


Fig. 6. Photoionization cross section and asymmetry parameter as a function of the photon energy. Expansion center for the partial wave channels on the H atom. The basis includes the partial wave channels up to $l = 8$ but not the localized channel.

fects mainly the length-gauge results, because only its orbitals located essentially on the lithium are poorly reproduced by the H-centered waves.

5. Conclusions

The present calculations agree fairly well with the previous ones upon pure PSGTO bases and this provides a safe check on the accuracy of both calculations. A further check of the reliability of all these calculations is provided by their internal consistency: the results are very weakly dependent on the nucleus where the STOCOS center is placed and, at RPA level, are almost independent from the gauge employed for the transition operator. On the whole, it may be concluded that both the present GTO/STOCOS and the previous PSGTO bases provide l, m manifolds that are radially complete inside the molecular region for a system of

the LiH size. It is conceivable that the GTO/STOCOS bases should allow to extend the calculation of the differential cross sections to larger systems.

Acknowledgement

The authors are grateful to Dr. I. Cacelli for helpful discussions and collaboration in developing the codes employed.

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